Guided Capstone Project Report

Problem Statement

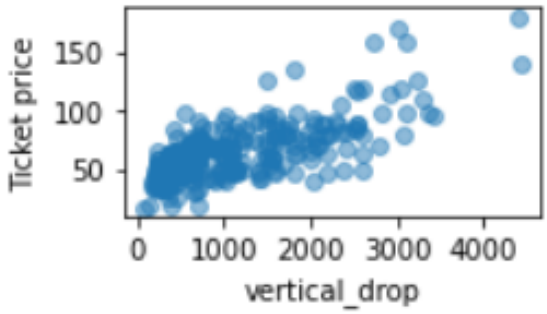
Big Mountain Resort aims to optimize its ticket pricing strategy to maintain a target profit margin of **9.2%**, while accounting for an additional **$1.54M in operational costs** for the upcoming season. The objective is to determine whether current pricing is sufficient or if operational or pricing adjustments are needed to sustain profitability.

Data Wrangling

The dataset was cleaned by handling missing values, removing irrelevant features, and ensuring consistency across variables. This step ensured the data was suitable for analysis and modeling.

Exploratory Data Analysis (EDA)

Exploratory analysis revealed strong relationships between resort features and ticket pricing. Resorts with greater vertical drop, more runs, and higher lift capacity generally charge higher prices. Big Mountain ranks highly across these features, indicating strong competitive positioning.



The relationship between vertical drop and ticket price shows a positive trend, indicating that resorts with greater vertical drop tend to charge higher prices.

Within the Montana market, Big Mountain already charges one of the highest ticket prices, suggesting limited room for pricing increases without additional value improvements.

Model Preprocessing & Feature Engineering

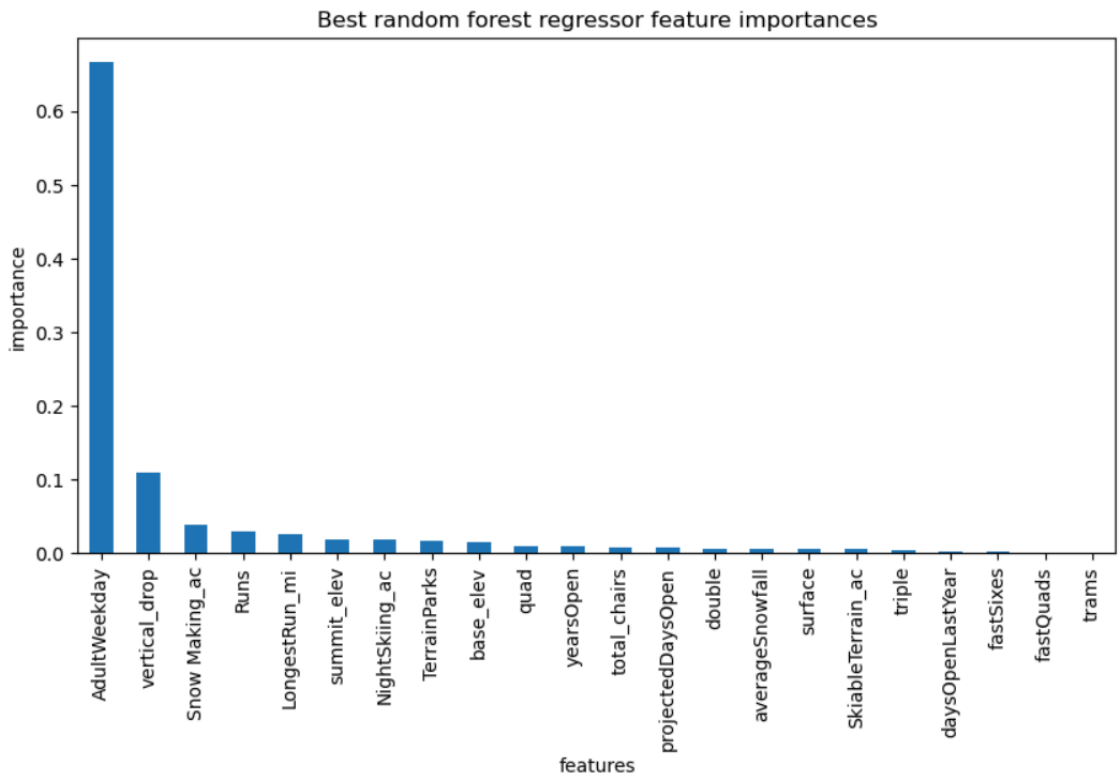
Preprocessing involved selecting relevant numerical features, handling missing values, and splitting the dataset into training and testing sets. Big Mountain was excluded from training to avoid bias in prediction.

Feature selection focused on variables directly influencing resort value, including terrain, lift infrastructure, and snowmaking capacity.

Modeling Approach

A Random Forest Regressor was used due to its ability to capture non-linear relationships between resort features and pricing.

Model performance was evaluated using R^2 and Mean Absolute Error (MAE), demonstrating strong predictive capability.



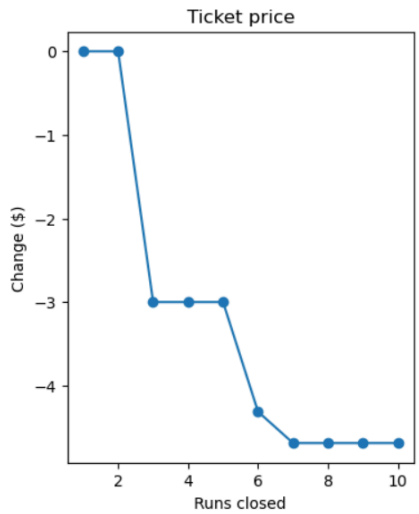
The feature importance plot highlights the key drivers of ticket pricing, with variables such as vertical drop and snowmaking capacity contributing significantly to the model's predictions.

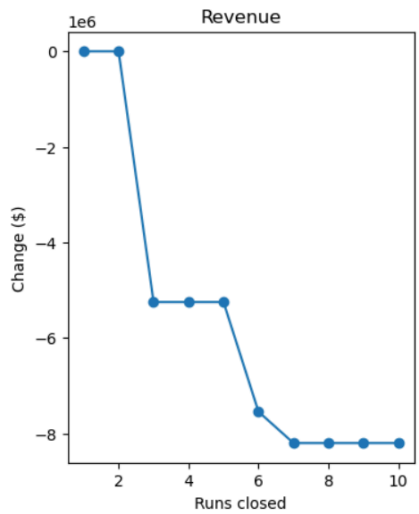
Scenario Modeling

The trained model was used to evaluate potential operational scenarios.

Scenario 1: Closing Runs

Closing runs led to a decrease in ticket price of up to approximately $4–5, with a corresponding decline in revenue.





As the number of closed runs increases, both ticket price and revenue decline significantly. This demonstrates that terrain availability is a key driver of value and that closing runs negatively impacts profitability.

Scenario 2 and 3: Infrastructure Improvements

Adding runs, vertical drop, lifts, and snowmaking resulted in a small decrease in predicted ticket price (approximately -$0.06 per ticket). When scaled across visitors, this leads to a noticeable reduction in seasonal revenue.

Pricing Recommendation

Big Mountain currently charges approximately $81 per ticket, which is already high relative to comparable resorts. The analysis suggests that increasing prices is not strongly supported by the model, as neither incremental infrastructure improvements nor operational changes provide sufficient justification for higher pricing. In particular, closing runs leads to a clear reduction in both ticket price and revenue, indicating that maintaining terrain availability is essential to preserving value.

Given the additional $1.54M in operating costs, it is unlikely that these costs can be recovered through price increases alone. Instead, the resort should maintain its current pricing strategy in the short term and focus on enhancing the overall customer experience to drive demand. Larger, more comprehensive improvements may be

considered, but they should be evaluated based on their ability to increase visitation and long-term revenue rather than relying solely on immediate pricing gains.

Conclusion

Big Mountain is competitively positioned and priced appropriately based on its features. Operational reductions negatively impact revenue, while incremental improvements do not significantly increase pricing potential. Maintaining profitability will require a broader strategy beyond pricing adjustments.

Future Scope of Work

This analysis was limited by the lack of detailed cost and demand data. While the dataset included ticket prices and resort features, it did not capture key financial metrics such as operating costs, maintenance expenses, or staffing requirements. Additionally, the absence of visitor demand and attendance data limits the ability to fully assess the relationship between pricing and total revenue.

Future work should focus on incorporating cost and profitability data to better evaluate the financial impact of operational decisions. Including demand forecasting and visitor behavior would also improve the model's ability to predict real-world outcomes. Finally, the model could be deployed as an interactive tool, such as a dashboard or web application, allowing business analysts to test different scenarios and explore pricing strategies without requiring direct involvement from the data science team.